



Severe anemia, anorexia, and uremia associated with diabetic foot infections: A case series

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ABSTRACT

Anemia of inflammation, as found in many chronic disease states, is common among persons with diabetic foot infections but is typically mild and self-limited. Herein we present four cases of patients with foot infections accompanied by severe anemia (nadir hemoglobin <8 gm/dL and resulting in transfusion of 4 + units of blood) as well as significant weight loss (30 + pounds), hyponatremia (<135 mmol/L), hypoalbuminemia (nadir <2 gm/dL), uremia and other metabolic derangements

1. Introduction

Despite being present in 67% of persons with foot ulcers [1] and 90% of persons undergoing leg amputation [2], the management of anemia of inflammation (previously referred to as anemia of chronic disease) in persons with foot ulcers or foot infections has not been well described. This may be because anemia of inflammation is typically characterized as "non-severe and not progressive", with hemoglobin levels ranging from 7 to 12 gm/dL [3], plateauing by three months from onset, and rarely dropping below 9 gm/dL [4].

In contrast, sepsis syndrome, first described nearly four decades ago [5], is recognized as an important clinical state of deterioration with a high associated mortality rate. Between the extremes of the potentially-fatal sepsis syndrome and seemingly-benign anemia of inflammation, few other systemic complications of foot infections had been reported until the recent characterizations of acute kidney injury [6], acute cardiac events [7] and more severe anemias [8].

Herein we describe four cases of severe anemia accompanied by extreme metabolic derangements occurring with foot infections. These four patients had similar combination of metabolic derangements that are different from other infectious complications previously described. Indeed, these cases may provide some rare characterization of the impact of chronic bacterial infection on whole body metabolism.

2. Case 1

A 35-year-old man with a 12-year history of uncontrolled type II diabetes presented to the hospital with a three month history of a left lateral midfoot ulcer and new onset of foot swelling, chest congestion, and fevers. He had initially experienced a 30 pound (14 kg; body-mass index 35 kg/m²) weight loss over the preceding month. He was admitted with hyperglycemia (299 mg/dL), leukocytosis, anemia (hemoglobin of 9.8 gm/dL), and acute kidney injury (2.29 mg/dL, from prior baseline of 1.05 mg/dL). He underwent operative drainage of an abscess in the soft tissue overlying the proximal fifth metatarsal and cuboid bone. Operative bone cultures showed *Streptococcus mitis*, methicillin-resistant *Staphylococcus aureus*, *Streptococcus agalactiae*, and *Actinomyces odontolyticus*. Operative fluid cultures identified those organisms as well as *Corynebacterium* and *Prevotella bivia*. He was discharged with oral amoxicillin-clavulonic acid.

He returned five weeks later with new onset swelling over the lateral malleolus, fatigue, fevers, anorexia, and frequent emesis after meals. He had lost an additional 10 pounds (4.5 kg; body-mass index of 33 kg/m²) and had been unable to take his oral antibiotics because of nausea and emesis. At the time of re-admission, he had fever (102.7 F), tachycardia (106–121/min), a relative hypotension (102/62 mmHg), acute kidney injury (creatinine 1.98 mg/dL), and a hemoglobin of 5.2 gm/dL. He had abnormal prothrombin time and partial thromboplastin time (Table 1)

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Table 1

Comparison of patient characteristics, including laboratory values and limb outcome.

Characteristic	Normal values	Case 1	Case 2	Case 3	Case 4
Baseline characteristics					
Age at presentation	N/A	35	68	75	47
Foot ulcer location(s)	None	Lateral, overlying proximal fifth metatarsal	Plantar aspect overlying fifth metatarsal head	Distal first toe	Dorsal foot and fifth toe
Foot ulcer dimensions at initial presentation, mm	0	25 × 20	25 × 19	8 × 8	10 × 10
Foot ulcer tissue depth	No ulcer	Tendon sheath	Dermis	Bone	Dermis
Lower extremity peripheral artery disease	No	No	Yes	Yes	No
Inflammatory & infection parameters					
Temperature at admission, °Fahrenheit (°Celsius)	97.0–100.4 (36.1–38.0)	101.2 (38.4)	100.4 (38.0)	99.0 (37.2)	99.7 (37.6)
Heart rate at admission, beats/min	< 100	123	149	104	111
Blood pressure at admission, mmHg	> 90	104/53	105/66	114/71	137/77
Respiratory rate, per minute	12–20	20	23	22	20
White blood cell count maximum, per mL	3.5–10.0	26.0	29.6	14.3	20.5
Erythrocyte sedimentation rate, mm/hr	< 20	> 100	> 100	86	> 100
C-reactive protein, mg/dL	< 0.7	23.45	3.56	17.6	2.98
Procalcitonin at presentation, ng/mL	< 0.09	31.97	2.44	0.10	–
Lactic acid at presentation, mmol/L	0.5–2.2	1.1	1.5	2.6	–
Blood cultures	Negative	Negative	Negative	Positive	Positive
Hemoglobin & iron parameters					
Hemoglobin baseline, gm/dL	12.0–18.0	–	13.5	13.0	13.5
Hemoglobin at presentation, gm/dL	12.0–18.0	7.0	6.7	7.6	7.8
Hemoglobin nadir, gm/dL	12.0–18.0	5.2	6.3	6.3	7.7
Hemoglobin most recent, gm/dL	12.0–18.0	10.7	7.3	11.1	8.8
Mean corpuscle volume, fL	81–100	82.8	85.4	79.7	80.1
Mean corpuscular hemoglobin concentration, gm/dL	33.0–36.0	32.5	33.9	33.1	32.9
Red cell distribution width, %	12.3–16.4	15.8	16.4	19.9	12.8
Total iron binding capacity, mcg/dL	260–445	110	147	140	–
Iron saturation	20–50	18.2	7.5	20.8	–
Serum iron, mcg/dL	28–182	20	11	18	–
Ferritin, ng/mL	11–336	2028	1172.1	282.4	237.7
Transferrin	180–382	79	105	< 70	–
Electrophoresis	No Spikes	No spikes	–	No spikes	–
RBC smear abnormalities	No	No	No	No	–
Other hematologic parameters					
Platelets, per mL	150–450	492	629	319	429
Reticulocytes, %	0.5–2.0	1.22	1.40	–	–
Fibrinogen, mg/dL	200–436	466	1112	668	–
D-dimer, mcg/mL	0–0.47	3.49	1.39	3.90	–
Lactic acid dehydrogenase, U/L	108–208	211	110	195	–
Haptoglobin, mg/dL	36–195	287	–	360	–
Protime maximum, seconds	12.0–14.7	19.7	18.9	17.4	17.7
Partial thromboplastin time maximum, seconds	23.6–33.6	58.0	76.2	45.1	–
Other anemia evaluations					
Serum folate, ng/mL	5.9–24.8	4.3	10.6	–	–
Vitamin B12, ng/mL	180–914	56	365	–	1096
Hepatitis panel	Negative	Negative	HBV negative	Negative	–
Serum renal parameters					
Serum creatinine at baseline, mg/dL	0.6–1.3	1.05	0.87	0.72	1.5
Serum creatinine maximum, mg/dL	0.6–1.3	3.45	1.32	1.52	3.42
Increase in creatinine, %	0	218	152	111	228
Acute kidney injury	No	Yes	No	Yes	Yes
Blood urea nitrogen maximum, mg/dL	7–19	39	30	29	32
Serum sodium nadir, mmol/L	136–145	125	132	131	134
Urinalysis parameters					
Urine color	Yellow or clear	Amber	Amber	Amber	Amber
Urine blood per dipstick	Negative	Moderate	Negative	Negative	Negative
Urine RBCs, per high power field	None	–	5	3	11
Urine hyaline casts per low power field	None	–	2–5	2–5	10–20
Urine protein, mg/dL	Negative	100	100	30	> 500
Glucose & metabolic parameters					
Glucose at presentation	< 105	220	266	95	162
HgbA1c, %	4.2–5.8	15.0	8.4	11.4	11.8
Body-mass index at baseline, kg·m ²	18.5–24.9	38.7	29.7	25.7	36.3
Weight loss, pounds (kg)	0	-59 (-27)	-55 (25)	-30 (14)	-57 (26)
Diarrhea history	No	No	No	No	No
Nadir Albumin, gm/dL	3.5–4.9	1.6	1.9	1.7	1.6
Ammonia, umol/L	10–35	–	23	19	–
Clinical outcomes					
Number of transfusions	0	7	5	5	4
Number of red cell units transfused	0	16	7	5	2

(continued on next page)

but no clinical signs of coagulopathy.

Transfusion of four units of red blood cells resulted in a repeat hemoglobin levels ranging from 6.3 to 7.6 gm/dL. Transfusion of an additional five units of red blood cells (for a total of nine units) during hospital days 2 and 3 resulted in a hemoglobin of 9.3 gm/dL (Fig. 1). Further operative drainage of abscess tracking toward the ankle was done on hospital day 2. Like before, operative fluid cultures at this time grew methicillin-resistant *Staphylococcus aureus*, *Streptococcus agalactiae*, and *Corynebacterium* but now grew *Pseudomonas aeruginosa*, *Enterococcus faecalis*, and *Prevotella oris*. Blood cultures were negative.

Given the severity of anemia and inadequate response to transfusions, an extensive anemia workup was done. Laboratory testing did not support significant iron deficiency and gastrointestinal investigation with Tc99m pertechnetate-tagged red blood cell scan, and upper endoscopy and colonoscopy did not reveal a source of blood loss. Peripheral smear did not show red blood cell fragmentation or other evidence of hemolysis. Laboratory analysis did not support diffuse intravascular coagulation, or other types of intravascular hemolysis, based upon normal haptoglobin and LDH, elevated fibrinogen, and low reticulocytes count. Testing for hepatitis B, hepatitis C, and human immunodeficiency virus was negative. His hemoglobin was 8.9 gm/dL at the time he was discharged from the hospital with an oral regimen of amoxicillin-clavulonic acid and levofloxacin.

The patient was readmitted again one month later for recrudescence of sepsis: persistent anorexia, fever (101.6 F), tachycardia (132/min), acute kidney injury (creatinine 1.76 mg/dL), anemia (7.0 gm/dL). He had lost an additional 20 pounds (9 kg) for a total of nearly 60 pounds (27 kg) since the onset of infection. CT of the lower extremity revealed air-density cavities throughout the ankle joint, strongly suggestive of septic arthritis. He underwent a two-stage transtibial amputation. During this hospitalization he received a total of seven units of blood for a hemoglobin levels ranging from 5.2 to 7.6 gm/dL, resulting in a final hemoglobin of 9.0 gm/dL. Repeat anemia evaluation done during this hospitalization revealed similar results as the previous hospitalization.

He improved over the subsequent year of recovery. He has gained 30 pounds (14 kg) of weight. His anemia is stable at 10.7 gm/dL, and his renal function is also stable at 1.97 mg/dL. His transtibial amputation healed, and he successfully exercises two hours a day with a limb prosthesis.

3. Case 2

A 68-year-old man with a 17-year history of type II diabetes initially presented to the hospital with chills, dizziness, gait imbalance, and gangrene of the right fifth toe. He had no fever, tachycardia, or hypotension. His hemoglobin at that time was 11.0 gm/dL, and his serum creatine was normal. He underwent amputation of this toe. Operative cultures of bone and soft tissue demonstrated *Enterococcus faecalis*, *Staphylococcus haemolyticus*, and *Corynebacterium*, and he was discharged home with oral doxycycline.

He was subsequently seen several times in the clinic. During these visits he complained of progressive anorexia, lightheadedness, chills, intermittent delirium, foot swelling, and purulent discharge. He declined hospital admission twice because of concern for acquiring COVID before eventually being re-admitted approximately one month after the toe amputation. At admission, he was noted to have severe sinus tachycardia (range 121–149/min), tachypnea, and anemia (hemoglobin 6.7 gm/dL). He had lost 30 pounds since his initial presentation. Foot x-ray revealed osteomyelitis now involving the fourth phalanx

and metatarsal bone. He underwent popliteal to dorsalis pedis bypass with an estimated 200 mL of blood loss, followed by three additional operations to drain an extensive abscess tracking along the plantar forefoot and midfoot, resect gangrenous toes, and address metatarsal osteomyelitis. He received seven units of red blood cell transfusions for his severe anemia (Fig. 1). Tissue and bone cultures from these operations grew *Proteus mirabilis*, *Klebsiella pneumoniae*, *Enterococcus faecalis*, *Streptococcus anginosus*, *Prevotella bivia*, and *Candida parapsilosis*.

A completion transmetatarsal amputation was done approximately two months after the leg bypass. Now four months since the leg bypass, his delirium has resolved. His dizziness has improved. He has lost an additional 25 pounds, and his hemoglobin remains low (7.3 gm/dL).

4. Case 3

A 75-year-old man with a 31-year-history of type II diabetes and history of a contralateral leg bypass for non-healing foot ulcer six years prior was initially admitted to the hospital with respiratory failure, *Klebsiella pneumoniae* bacteremia, and an eschar at the tip of the left hallux. He underwent left leg bypass and partial hallux amputation for gangrene and abscess. Operative cultures from that operation grew *Staphylococcus epidermidis*, *Staphylococcus simulans*, and *Corynebacterium*. He had one urinalysis that showed 644 red blood cells per high power field and > 500 mg/dL protein, but two others within the same week that had shown 2 and 23 red blood cells per high power field and 30 and 100 mg/dL protein, respectively.

After returning to the hospital one month later with weakness, hypotension (86/53 mmHg), tachycardia (110–136/min), fever (102.2 F), and acute kidney injury (creatinine 3.96 mg/dL), he underwent additional (unplanned) operative drainage of infection due to *Enterococcus faecalis* and *Candida parapsilosis* at the toe amputation site. He was transfused two units of red blood cells and discharged with oral amoxicillin-clavulonic acid and fluconazole.

He was then re-admitted to the hospital another two months later. By this time, he had experienced weight loss of more than 55 pounds since the leg bypass was performed. His delirium had worsened, and his hemoglobin had decreased progressively from 13.0 to 7.6 gm/dL. Without another clear etiology identified, recommendations for palliative wound care of the open toe amputation site transitioned to a recommendation for transtibial amputation of the leg. Pathology from this operation demonstrated osteomyelitis in the forefoot. He was transfused three units of red blood cells. The delirium was slow to improve during early in-hospital recovery but normalized – along with his kidney function – in later follow-up. Now two years since leg amputation, he has regained 25 pounds of weight, and his most recent hemoglobin is 11.1 gm/dL.

5. Case 4

A 47-year-old man with a 23-year-history of type II diabetes and left foot ulcer initially presented to his primary care provider with malaise, anorexia, mild generalized abdominal pains, constipation, and an initial 20 pound weight loss (body mass index of 36 kg/m²). His creatinine had increased from a normal baseline to 2.5 mg/dL. He had a normal hemoglobin (13.5 gm/dL). He received wound care from a podiatrist. He had not been prescribed antibiotics.

He then presented to the hospital about a month later after developing foot edema, serosanguinous fluid, and foul odor. He had lost an additional 37 pounds of body weight since his previous presentation. He had fever to 100.7 F, tachycardia (104–120/min), leukocytosis (19/

Table 1 (continued)

Characteristic	Normal values	Case 1	Case 2	Case 3	Case 4
Leg amputation	No	Yes	No	Yes	No
Alive?	Yes	Yes	Yes	Yes	No
Follow-up time	N/A	10 months	4 months	28 months	30 months

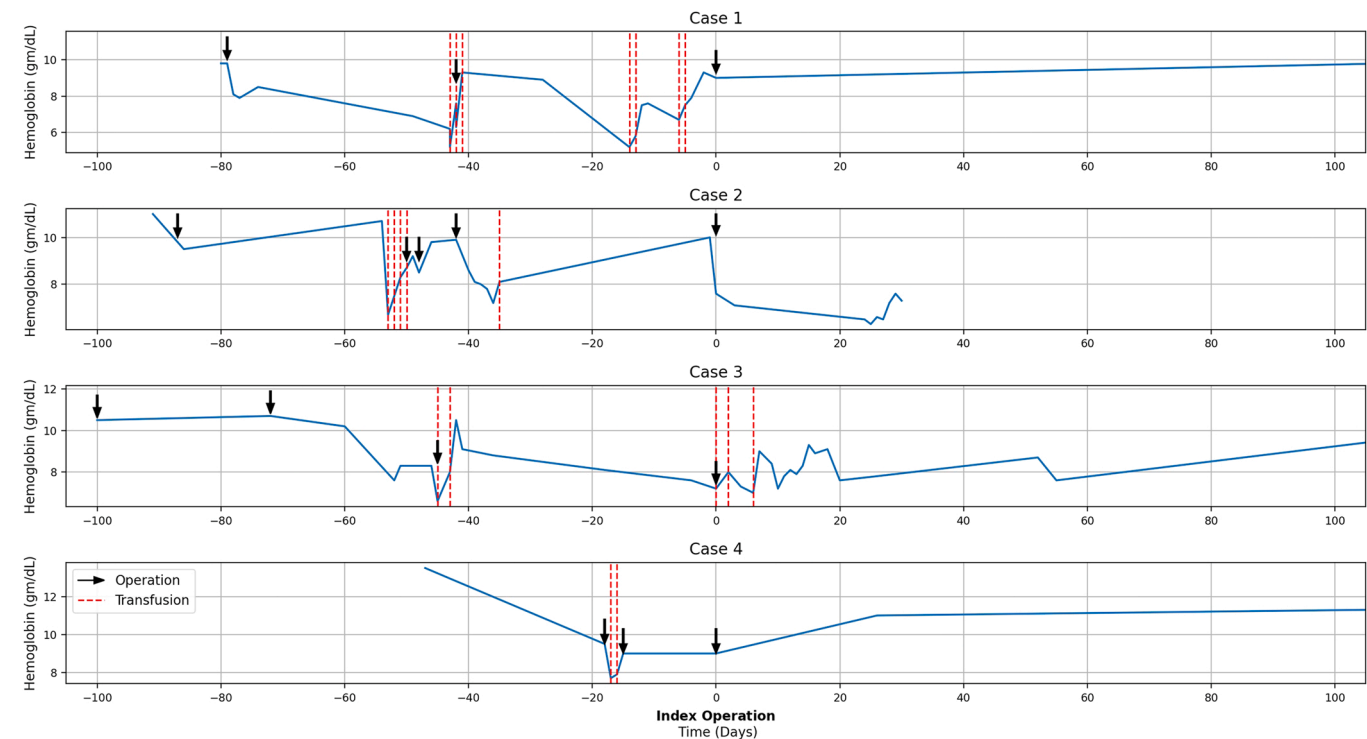


Fig. 1. Hemoglobin levels over time for the four patients described here. Arrows represent operations, with time 0 defined by the index operation for foot infection. Dotted red vertical lines represent red cell transfusions. Notable are the rapid decreases to very low nadir hemoglobin levels, also improvement in three of the patients (though not back to baseline levels).

mL), hyperglycemic in the 400 s, and acute kidney injury (3.4 mg/dL). His hemoglobin level was 9.5 gm/dL on admission. There was a 1 cm diameter ulcer in the center of the plantar midfoot with severe focal soft tissue swelling surrounding this area and in the dorsum of the foot. Foot x-ray demonstrated extensive dorsal subcutaneous gas and possible osteomyelitis at the head of the second metatarsal.

Operative drainage of a foot abscess was performed, and tissue cultures grew group B *Streptococcus*, *Citrobacter diversus*, alpha-hemolytic streptococci (identified only as not *pneumoniae*, not *Enterococcus*), and *Bacteroides fragilis*. Blood cultures drawn at the time of admission eventually also demonstrated *Bacteroides fragilis*. He received two units of red blood cell transfusion for a nadir hemoglobin of 7.7 gm/dL, and he underwent an additional operative debridement four days after the first. His foot wound was surgically closed after second toe amputation done about three weeks after his hospitalization, and it remained healed throughout follow-up. Bone histopathology from this operation demonstrated osteomyelitis. His hemoglobin peaked at 12.2 gm/dL approximately one year after his foot infection presentation, then ranged from 8.8 to 10.8 gm/dL during subsequent follow-up. He died after sudden cardiac arrest approximately 2.5 years after his presentation for foot infection.

6. Discussion

The four cases reported here detail extreme degrees of anemia, anorexia and kidney injury that are rare but notable metabolic derangements not previously described. Different from the ignition-and-blaze that septic shock from acute infections such as COVID-19 or H1N1 influenza can represent, these metabolic derangements have a time course more akin to embers kindling a pyre: steady, spreading, and pernicious. These cases, through representing a very small fraction of the foot infections seen at our hospital, may provide further insights into the impact of infection on whole body metabolism.

Similar to anemia of inflammation, we often see decreased appetite and weight loss in patients with foot infection. Like anemia of

inflammation, it is almost always mild. The weight loss is typically in the range of 5–10 pounds [4]. The severe weight loss described in here, ranging from 30 to nearly 60 pounds, seems as extreme as the degree of anemia and is probably, by some mechanisms we cannot elucidate, an overlapping phenomenon. The weight loss does not seem to be a complication of oral antibiotics, as only two patients had consistently taken oral antibiotics and none had diarrhea, nausea, PO intolerance, or comorbidities causing anorexia.

It is unclear if these cases represent far-advanced but typical metabolic consequences of severe infection, or if they represent an unusual syndrome occurring in a subset of patients with foot infections. We have noted that, in general, younger patients with foot infection seem to more often manifest criteria of the systemic inflammatory response syndrome, an observation also described in intraabdominal infections [9]. Two of our cases did occur in younger patients (35 and 47 years old), but two others occurred in elderly patients (68 and 75 years old) more typical of the diabetic foot ulcer population at our hospital.

All four patients had foot soft tissue infections and osteomyelitis

Table 2
Consistencies in patients described in this series.

PRESENT in ALL cases	ABSENT in ALL cases
Type II diabetes mellitus	Source of blood loss
Tachycardia & tachypnea	Evidence of hemolysis
Severe anemia (Hgb<8) leading to red cell transfusions	Absolute thrombocytopenia
Polymicrobial soft tissue and bone infection of the foot that include beta-hemolytic streptococcal species	Necrotizing soft tissue infection
Intractable nausea, anorexia, and emesis leading to 30 + lbs. weight loss	Septic shock
Uremia, amber-colored urine with proteinuria	Diarrhea
Leukocytosis, elevated c-reactive protein, elevated erythrocyte sedimentation rate	Clinical signs of coagulopathy
Hyponatremia and severe hypoalbuminemia (<2 gm/dL)	
Elevated protime	

(Table 2). In two cases, some degree of infection was recognized as being present for at least a month (though the other two likely also some degree of infection present prior to admission). All had polymicrobial infections that included beta-hemolytic *Streptococci*. Beyond these, few other features seem consistent. The presence of *Corynebacterium*, *Candida*, and gram negative organisms were common but not uniform. Only one had bacteremia with an organism identified in the foot (another with bacteremia attributed to bacterial pneumonia). Three patients had poor glucose control (hemoglobin A1c >11%) prior to presentation, but not all (hemoglobin A1c of 8.4% in patient #2). Two were obese and two were overweight. Delirium and lower extremity peripheral artery disease were present in both of the elderly patients but neither of the younger patients.

In all of the patients, inflammation was a likely significant driver of their anemia, with no other plausible explanation to account for the severity of the anemia. None of the patients had significant hemorrhage either related to the surgical site or gastrointestinal tract. None had evidence of extravascular or intravascular hemolysis, including diffuse intravascular coagulation. In one case, there was evidence of concomitant iron deficiency for which complete evaluation has not been completed. These patients had presentations that shared some but not all of the features of hemolytic-uremic syndrome. Anemia, elevated creatinine, and anorexia are present in hemolytic anemia syndrome, but notably absent were thrombocytopenia (platelet counts that were notably elevated in three of the four, and decreased by 68% in the fourth but within normal range), diarrhea, decreased haptoglobin (slightly elevated in the two patients with levels measured), and elevated lactated dehydrogenase levels (normal in three of the four patients in which it was measured).

Anemia of inflammation, primarily a disorder of iron distribution, is mediated by high levels of cytokines such as TNF-alpha, interferon-gamma, interleukin-1, and interleukin-6 [3,10]. In particular, interleukin-6 stimulates and increases hepcidin levels, which then restrict the iron exporting activity of ferroportin [3,10]. By interfering with iron absorption and mobilization from macrophages and hepatocytes, the interaction among these cytokines disrupts iron homeostasis and reduces the amount of circulating iron available for hemoglobin formation. In addition to systemic inflammation, decreased renal function contributes to decreased hepcidin clearance and further loss of appropriate erythropoietin production. Anemia has thus been noted to be twice as common in persons with diabetes compared to those without [11–13]. Beta-hemolytic streptococci such as *Enterococcus faecalis* was also seen in all four patients. This species of bacteria produces streptolysin, an exotoxin that explains the ability of these organisms to lyse red blood cells in vitro (as seen in the urinalysis in patient #3). Since a 1934 description of anemia, in which amber urine and death in mice were

seen after injection of streptococcal "hemotoxin" in mice [14], no clinical reports have clearly linked streptococcal infection with any of the metabolic derangements described herein.

Further study is needed to understand the pathogenesis of these clinical presentations and to understand why they occur in a small subset of patients with foot ulcers. Unless another cause is identified, patients with foot ulcers who have anemia, anorexia, and acute kidney injury should be presumed to have a foot infection – even if typical signs of infection such as erythema, swelling, and fluctuance are absent. Furthermore, noting the clear relationship seen on this series, we suggest these parameters can and should be used to track the progress in treating infection. Whether they represent unusual manifestations of sepsis pathophysiology or a rare, separate infection-related syndrome, these four cases may help sound the depths of the clinical host-pathogen interactions.

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